# ABDELRAHMAN MOHAMED AHMED

**Software Engineer**

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LinkedIn | GitHub | Portfolio

## PROFESSIONAL SUMMARY

Driven by a passion for building scalable, cloud-native systems, with 2 years of Python experience and hands-on expertise in Java/Spring Boot, FastAPI and React/TypeScript. Proficient in distributed-systems patterns (Kafka, sagas, outbox), containerization and orchestration (Docker, Kubernetes, Terraform), CI/CD automation and observability. Experienced in chatbot workflow automation and in teaching Python. Seeking a Backend and DevOps or Full-Stack role to gain industry experience related to the field, collaborate effectively within a team, and grow into high-impact positions.

## EDUCATION

**Modern Academy for Engineering and Technology** 2021 – 2026

Faculty: Computer Engineering

Bachelor of Computer Engineering

## EXPERIENCE

**IEEE** — Python Instructor & Vice-Head 23 Mar 2026 – 5 May 2026

- Taught the Python programming language as an instructor.
- Held the vice-head position.

**OctoBot** — Chatbot Workflow Builder 12 Dec 2025 – 5 Mar 2026

- Built the workflows behind chatbots at a company specializing in automation and chatbot development.
- Contributed to the company's automation solutions through chatbot workflow development.

## PROJECTS

**Event Management Platform**

- Built event, ticketing, booking, payment and notification services as Java/Spring Boot microservices with a React/TypeScript frontend and JWT security.
- Applied the outbox pattern, Kafka-driven notifications and a payment/refund saga; added retry and dead-letter handling.
- Instrumented all services with Prometheus, Grafana, Loki, OpenTelemetry and Jaeger; containerized with Docker Compose.

**AI Project Manager Agent**

- Designed an AI planning platform that turns project ideas into plans, milestones, tasks, schedules, dependency graphs and risk views.
- Engineered a FastAPI/Python backend with background workers (job leasing, heartbeats, retries, cancellation) and provider-neutral AI adapters with validation and repair loops.
- Built a React/TypeScript/Vite frontend with review-and-approval workflows, plus a packaged Windows one-click launcher.

**Banking Transaction System**

- Built a cloud-native TypeScript banking system with account management, transfers, ledger, fraud detection and notifications as independent microservices.
- Designed the infrastructure with Docker Compose, Kubernetes, Terraform and GitHub Actions CI/CD, backed by observability tooling.

**Cloud-Native E-Commerce Platform**

- Developed a microservices platform with a React/Vite frontend, Nginx API gateway, and Node.js and Python/FastAPI services using Redis and PostgreSQL.
- Deployed to Kubernetes with autoscaling and blue-green deployments on AWS/EKS via Terraform; added CI/CD and Prometheus/Grafana monitoring.

**TWSELA** — Courier & Delivery Management Platform

- Contributed to an enterprise courier platform for MENA logistics with Arabic-first RTL support across five roles.
- Worked on a Spring Boot 3.3.3 / Java 17 backend (Spring Security, JWT, JPA, Redis, WebSockets, Resilience4j) covering shipments, courier assignment, proof of delivery and COD reconciliation.
- Built React 18/Vite/Tailwind features, including real-time updates and Recharts analytics dashboards; authored security, database and roadmap documentation.

**Aircraft Management, Operation & Maintenance Using AI** — DevOps

- Built GitHub Actions CI/CD pipelines with Docker image builds, GHCR publishing and Trivy/tfsec scanning across a Java/TypeScript/Python stack.
- Managed Kubernetes and AWS/EKS infrastructure with Terraform and Kustomize, resolving crash-loop, database privilege and ingress issues.
- Strengthened quality through Playwright/Vitest test auditing and health checks.

**StockPilot** — Inventory & Sales Operations Platform

- Contributed to a platform with a NestJS backend, a separate frontend and a Python ML service.
- Built Dockerfiles, multi-service Docker Compose environments and GitHub Actions CI/CD with build gates and health checks.

**Academic Projects**

- **Insurance Company Software System and Database:** desktop-style policy system with an ERD-modeled, normalized relational database.
- **PCM Simulation (MATLAB/Simulink):** sampling, quantization and encoding of analog signals.
- **Cybersecurity Hotel Network:** firewalls, segmentation and access control.
- **Machine Learning Flower Classification:** dataset preparation, feature extraction, training and evaluation.
- **DevOps QR Code Scanner:** Docker-based CI/CD pipeline that halts safely on failure.

## SKILLS

**Technical**

- **Languages:** Python (2 years), Java, TypeScript, JavaScript, C
- **Backend:** Spring Boot, FastAPI, NestJS, Node.js/Express
- **Frontend:** React, Vite, Tailwind CSS
- **Data and Messaging:** PostgreSQL, MySQL, Redis, Apache Kafka
- **Cloud and DevOps:** Docker, Kubernetes, Terraform, GitHub Actions, AWS/EKS
- **Observability:** Prometheus, Grafana, OpenTelemetry

**Soft**

- Public Speaking
- Communication
- Teaching
- Leadership
- Teamwork
- Cross-Team Collaboration
- Problem-Solving
- Adaptability
- Time Management
- Critical Thinking

## LANGUAGES

Arabic (Primary), English (B2)

## REFERENCES

References are available upon request.